

# Keynes's Theory of Money

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# 1 Keynes's Theory of Money

## 1.1 Introduction

John Maynard Keynes revolutionized monetary economics by examining the roles money plays in the economy and their macroeconomic implications. Writing during the Great Depression, Keynes challenged the classical view that money is merely a neutral medium of exchange or “veil” that affects only prices but not real output or employment. In classical theory, the economy was thought to naturally gravitate to full employment equilibrium via flexible prices (the classical dichotomy), implying money had no real effects in the long run. Keynes, however, argued that in a monetary economy – where money actively influences decisions – there is no automatic guarantee of full employment. He developed a theory of liquidity preference and money demand that highlights how money can matter in both the short run and long run. This chapter examines Keynes's theory of money demand (transactions, precautionary, and speculative motives), his concept of the liquidity trap, and the resulting disequilibrium analysis using the IS-LM model. We also contrast the neutrality of money in classical theory with Keynes's view of non-neutrality, and discuss the Pigou effect as a classical critique of Keynes. Throughout, we integrate historical and contemporary examples – from Japan's 1990s stagnation to post-2008 low interest environments in the US, UK, and Eurozone – to illustrate these concepts in practice. ([Keynes, 1936](#); [Lewis and Mizen, 2000](#))

## 1.2 The Exceptional Nature of the Money Market

### 1.2.1 Why Money-Market Disequilibrium Differs

Keynes's key analytical move is to treat the money market as exceptional. In many markets, an excess demand triggers adjustment through at least three channels: (i) the good's own price rises, (ii) supply expands, and (iii) agents substitute toward close substitutes. Keynes argued that these equilibrating forces are weak or absent for money in its role as a pure medium of exchange. ([Keynes, 1936](#))

### 1.2.2 Three Special Features of Money

Keynes highlighted three features of the money market that make it hard to clear through “normal” market mechanisms: ([Keynes, 1936](#))

1. **The nominal price of money is unresponsive to disequilibrium** — Money has a face value. Unlike ordinary goods, an excess demand for money does not “bid up” the nominal value of money in the same way that excess demand for apples bids up apple prices.
2. **Supply is unresponsive because the money stock is exogenously determined** — The stock of money is set by the monetary authority rather than being produced in response to higher “money prices”.
3. **There are no close substitutes as media of exchange** — Even if near-money instruments exist today, Keynes’s emphasis is that money’s payment finality and transactional acceptability are difficult to replicate fully. This matters when confidence collapses: agents may prefer money to specific private liabilities (bank deposits, firm credits) perceived as risky.

These features underpin Keynes’s claim that money-market adjustment is weaker than in ordinary goods markets. ([Keynes, 1936](#))

### 1.2.3 A Modern Interpretation: Safe Assets and Flight-to-Quality

In modern financial systems, some substitutes exist (money market funds, repos, short-maturity government bills). However, Keynes’s core point survives: in stress episodes, investors demand the most liquid and safest instrument, and the system can experience a shortage of safe and liquid assets. This is consistent with observed “flight-to-quality” and “flight-to-liquidity” dynamics in financial crises. ([Lewis and Mizen, 2000](#))

## 1.3 Say’s Law, Walras’ Law, and Unemployment in a Monetary Economy

### 1.3.1 Keynes’s Mechanism: Saving as Hoarding

In a non-monetary (or effectively barter) economy, an increase in saving typically reallocates demand rather than destroying it. Classical reasoning holds: higher saving means more purchases of durable goods or more lending to firms for investment, shifting production across sectors but leaving aggregate employment roughly unchanged. ([Lewis and Mizen, 2000](#))

In a monetary economy, however, higher desired saving can take the form of hoarding money (increasing money demand) rather than purchasing goods or financing investment. When this occurs: ([Keynes, 1936](#))

- Overall demand for goods falls,
- Demand for money rises,
- Labour demand falls,
- Unemployment emerges.

### 1.3.2 Walras' Law Does Not Imply Say's Law in a Monetary Economy

A crucial insight from the lecture notes: **in a monetary economy, Walras' Law no longer implies Say's Law**. An excess demand for money can co-exist with an excess supply of goods. When agents attempt to rebuild liquidity positions, desired spending falls below output. This is Keynes's fundamental explanation for why unemployment can be an equilibrium outcome in a monetary economy: the “missing demand” is not redirected automatically into other goods; it can be redirected into money balances. ([Walras, 1874](#); [Keynes, 1936](#))

## 1.4 Normal Mechanisms for Restoring Equilibrium (and Why They Can Fail)

### 1.4.1 Adjustment Through Other Markets

Keynes acknowledges that money-market disequilibrium can in principle be eliminated via adjustment in other markets, even if money's own “price” does not move: ([Keynes, 1936](#))

1. **Interest rate adjustment:** If returns on alternative assets rise, agents substitute away from money; money demand falls.
2. **Price-level adjustment:** If goods prices fall, real balances  $M/P$  rise and goods demand may increase; money demand (as a share of expenditure) falls.

This is crucial: Keynes is not claiming the money market is always permanently out of equilibrium. He is claiming that adjustment can be slow, incomplete, or blocked—especially when prices are sticky and when interest rates approach a floor. ([Keynes, 1936](#))

### 1.4.2 Price Stickiness as a Macroeconomic Amplifier

The key point is that adjustment mechanisms “might” work but do not necessarily “will” work because of price stickiness. Equilibrating adjustments may take significant time. Modern macroeconomic evidence is consistent with the idea that wages and prices (especially nominal wages and administered prices) adjust sluggishly, so output and employment take the hit in the interim. ([Lewis and Mizen, 2000](#))

## 1.5 Keynes's Theory of Money Demand

### 1.5.1 Classical vs. Keynesian Interest Theory

#### 1.5.1.1 Classical Theory: Loanable Funds

Classical economists view interest as the reward for saving (deferring consumption) and determine it via supply and demand for loanable funds: (Lewis and Mizen, 2000) Figure 1 shows the classical loanable funds market and the equilibrium interest rate implied by  $S = I$ .

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.4.3

```
# Data
lf <- seq(0, 10, length.out = 100)
S <- 2 + 0.5 * lf # Saving supply
I <- 7 - 0.5 * lf # Investment demand

# Equilibrium
eq_lf <- 5
eq_i <- 4.5

# Plot
ggplot() +
  geom_line(aes(x = lf, y = S, color = "S (Saving supply)"), linewidth = 1) +
  geom_line(aes(x = lf, y = I, color = "I (Investment demand)"), linewidth = 1) +
  annotate("point", x = eq_lf, y = eq_i, size = 3) +
  geom_segment(aes(x = eq_lf, xend = eq_lf, y = 0, yend = eq_i), linetype = "dashed", alpha = 0.5) +
  geom_segment(aes(x = 0, xend = eq_lf, y = eq_i, yend = eq_i), linetype = "dashed", alpha = 0.5) +
  annotate("text", x = eq_lf + 0.5, y = eq_i, label = "Equilibrium\n(i*)", size = 3.5) +
  scale_color_manual(values = c("S (Saving supply)" = "blue", "I (Investment demand)" = "red")) +
  labs(x = "Loanable Funds", y = "Interest Rate (i)",
       title = "Classical Loanable Funds Market", color = "") +
  theme_minimal() +
  theme(legend.position = "top")
```

## Classical Loanable Funds Market

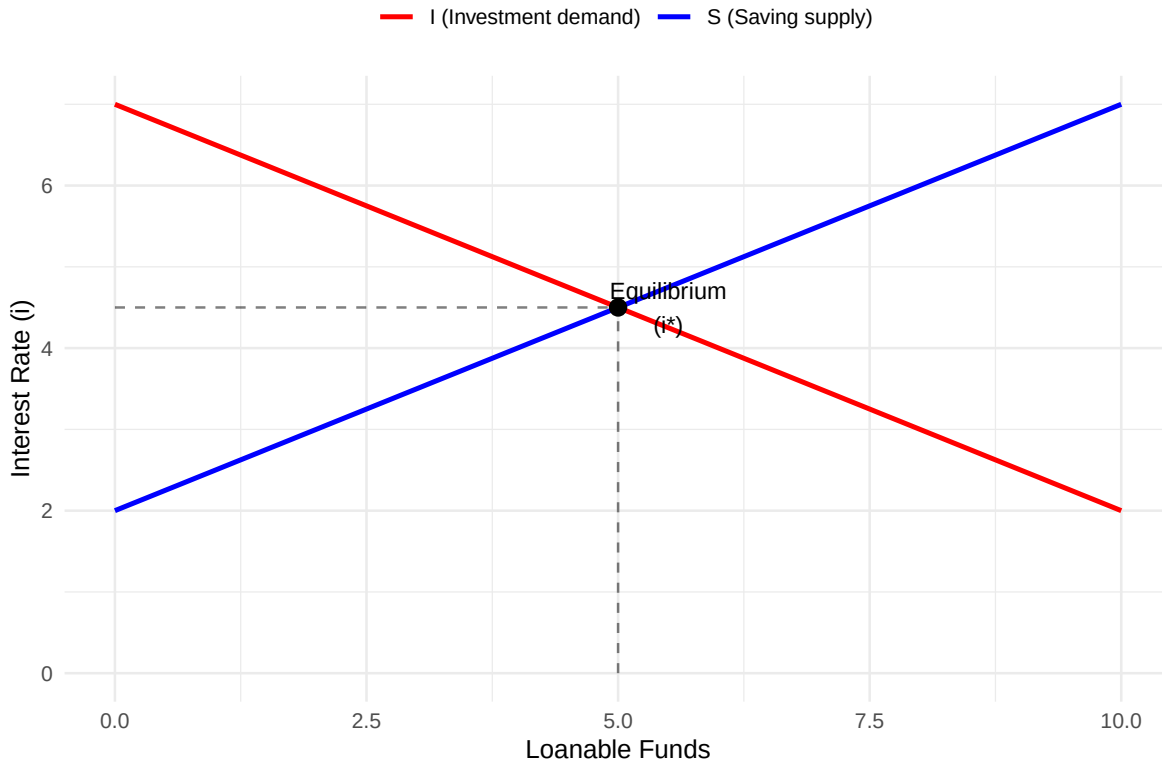


Figure 1: Classical loanable funds market and interest rate determination

### 1.5.1.2 Keynes's Liquidity Preference: Core View

Keynes reframes interest as the reward for holding wealth in illiquid form rather than liquid form. Thus: (Keynes, 1936)

- A fall in  $i$  increases money demand (agents prefer liquidity over bonds),
- Money demand and money supply jointly determine  $i$ ,
- Higher income shifts money demand right and raises  $i$  (via higher transactions demand).

Keynes also notes the reconciliation with the classical view via a system approach (IS–LM), as later formalized by Hicks. (Hicks, 1937)

Classical economists explained interest rate determination through the loanable funds market, where the interest rate adjusts to equate saving and investment. In that view (often illustrated by a supply-of-savings and demand-for-investment diagram), the interest rate is the price of funds: households' willingness to save (defer consumption) increases with a higher interest return, while firms' willingness to invest in new capital decreases with higher financing costs.

Equilibrium occurs when  $S = I$ , and at that point the interest rate balances the economy's ex ante saving and investment. An increase in desired saving (e.g. due to higher income or thriftiness) would shift the savings supply curve rightward and drive the equilibrium interest rate down, encouraging more investment. In this classical view, interest is ultimately governed by real factors (productivity and time preference) and saving-investment behavior, not directly by the quantity of money. (Lewis and Mizen, 2000)

Keynes's liquidity preference theory, by contrast, emphasizes money market equilibrium. He did not deny that income affects saving and investment, but in the short run he assumed saving is mostly driven by income (via the marginal propensity to consume) rather than by the interest rate. Investment, on the other hand, depends on interest and profit expectations. In Keynes's theory, the interest rate adjusts until money demanded = money supplied, given the level of income and money stock. For example, if income rises (more transactions), the money demand curve  $L(Y, i)$  shifts right; if the money supply is unchanged, this excess demand for money will drive up interest rates until people are content holding the available money (because higher  $i$  reduces the speculative and precautionary demand). Conversely, if the central bank increases the money supply, initially there is excess money in portfolios, so people try to rebalance by buying bonds and assets – bidding up their prices and pushing down the interest rate. Thus, in liquidity preference theory, an increase in income raises interest rates (money demand up), and an increase in money supply lowers interest rates (money supply up), ceteris paribus. (Keynes, 1936)

Keynes's view can be reconciled with the classical loanable funds approach by considering interactions in both goods and money markets simultaneously. In fact, the IS-LM model (originally formulated by John Hicks in 1937 as an interpretation of Keynes) combines the two: the IS curve captures equilibrium in the goods market (where investment equals saving, or equivalently aggregate demand equals output), and the LM curve captures equilibrium in the money market (liquidity preference equals money supply). We will use IS-LM shortly to analyze Keynesian equilibrium and disequilibrium conditions. First, however, we examine Keynes's famous concept of the liquidity trap, an extreme case of liquidity preference with profound policy implications. (Keynes, 1936; Hicks, 1937)

### 1.5.2 The Three Motives for Holding Money

Keynes's theory of money demand begins with the recognition of three motives for holding money, expanding beyond the single transactions motive emphasized by classical economists. In classical thinking (e.g. quantity theory), people hold money only to facilitate transactions – the medium of exchange function – and any excess money would quickly be spent, rendering money neutral in its effect on the economy. Keynes acknowledged the transactions motive (holding money for routine purchases of goods and services) but also identified two additional motives arising from money's role as a liquid store of value that pays no interest: (Keynes, 1936)

- **Precautionary motive** – holding money for unexpected contingencies or emergencies that might require quick expenditure. Even households and firms with planned budgets keep some funds in readily accessible cash or deposits as a precaution against unforeseen events (medical bills, urgent repairs, etc.). This motive implies money demand increases with the uncertainty of the environment or the need for flexibility. (Keynes, 1936)

- **Speculative motive** – holding money instead of other assets when the return on bonds or other interest-bearing assets is low, or when those assets are expected to fall in price. This is related to expectations about future interest rates. If people expect interest rates to rise (which means bond prices will fall), they may prefer to hold wealth in money now and wait for a better opportunity to buy bonds later. Money is perfectly liquid (it can be used directly for transactions and does not incur capital loss), whereas bonds and other assets carry interest but can fluctuate in value. The speculative demand for money is inversely related to the interest rate: when interest ( $i$ ) is high, people are more willing to hold bonds (earning interest) and less inclined to hold idle money; but when  $i$  is low, the opportunity cost of holding money is small, so people are more inclined to keep wealth liquid in money form. (Keynes, 1936)

### 1.5.3 Liquidity Preference Function

Keynes formalized these ideas in his liquidity preference theory. The demand for money can be expressed as a function  $L(Y, i)$ , where  $Y$  is income and  $i$  is the interest rate. The transactions and precautionary motives imply that  $L$  increases with real income  $Y$  (higher income means more spending and a larger buffer for contingencies). The speculative motive implies that  $L$  decreases as  $i$  rises (higher interest provides a reward for parting with liquidity). In other words: (Keynes, 1936)

- $\frac{\partial L}{\partial Y} > 0$ : Higher income raises money demand (for transactions and safety reserves).
- $\frac{\partial L}{\partial i} < 0$ : Higher interest rates reduce money demanded (as the public holds interest-bearing assets instead).

Keynes described interest rate as the reward for relinquishing liquidity. Unlike classical theory which saw interest as the reward for saving (waiting to consume), Keynes argued that at any given level of income, the interest rate adjusts to equilibrate money demand with the fixed money supply. In equilibrium, money supply  $M$  = money demand  $L(Y, i)$ , and the interest rate is the “price” that brings this about. If the central bank sets a fixed nominal money supply  $M$  (and for a given price level  $P$ , a fixed real money supply  $M/P$ ), then the interest rate will move until money market clears. (Keynes, 1936)



## 1.6 The Liquidity Trap and Its Implications

### 1.6.1 Definition in Liquidity Preference Terms

At very low interest rates  $i_L$ , the speculative motive dominates: agents are reluctant to hold bonds because: (Keynes, 1936)

- Yields are too low to compensate for inconvenience and illiquidity,
- They expect rates cannot go much lower (so bond prices are “too high”) and thus fear capital losses when rates rise.

Hence the liquidity preference schedule becomes horizontal: money demand becomes infinitely interest-elastic. (Keynes, 1936) Figure 2 shows the liquidity preference schedule with the horizontal segment at the trap floor.

```
library(ggplot2)

# Money demand - normal region
M_normal <- seq(0, 8, length.out = 50)
i_normal <- 8 - M_normal

# Liquidity trap region
M_trap <- seq(8, 12, length.out = 20)
i_trap <- rep(1.5, length(M_trap))

# Combine
df <- data.frame(
  M = c(M_normal, M_trap),
  i = c(i_normal, i_trap)
)

# Plot
ggplot(df, aes(x = M, y = i)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_hline(yintercept = 1.5, color = "red",
    linetype = "dashed", linewidth = 0.8, alpha = 0.7) +
  annotate("rect", xmin = 8, xmax = 12, ymin = 0, ymax = 1.5,
    fill = "red", alpha = 0.15) +
  annotate("text", x = 10, y = 1.9, label = "i[L]~(trap~floor)",
    parse = TRUE, color = "red", fontface = "bold", size = 4) +
  annotate("rect", xmin = 1.5, xmax = 4.5, ymin = 4.5, ymax = 5.5,
    fill = "wheat", alpha = 0.7) +
  annotate("text", x = 3, y = 5,
```

```

    label = "Normal region:\ndownward sloping",
    size = 3.5) +
labs(x = "Money Demand (M)", y = "Interest Rate (i)",
    title = "Liquidity Preference Schedule") +
theme_minimal() +
coord_cartesian(xlim = c(0, 12), ylim = c(0, 9))

```

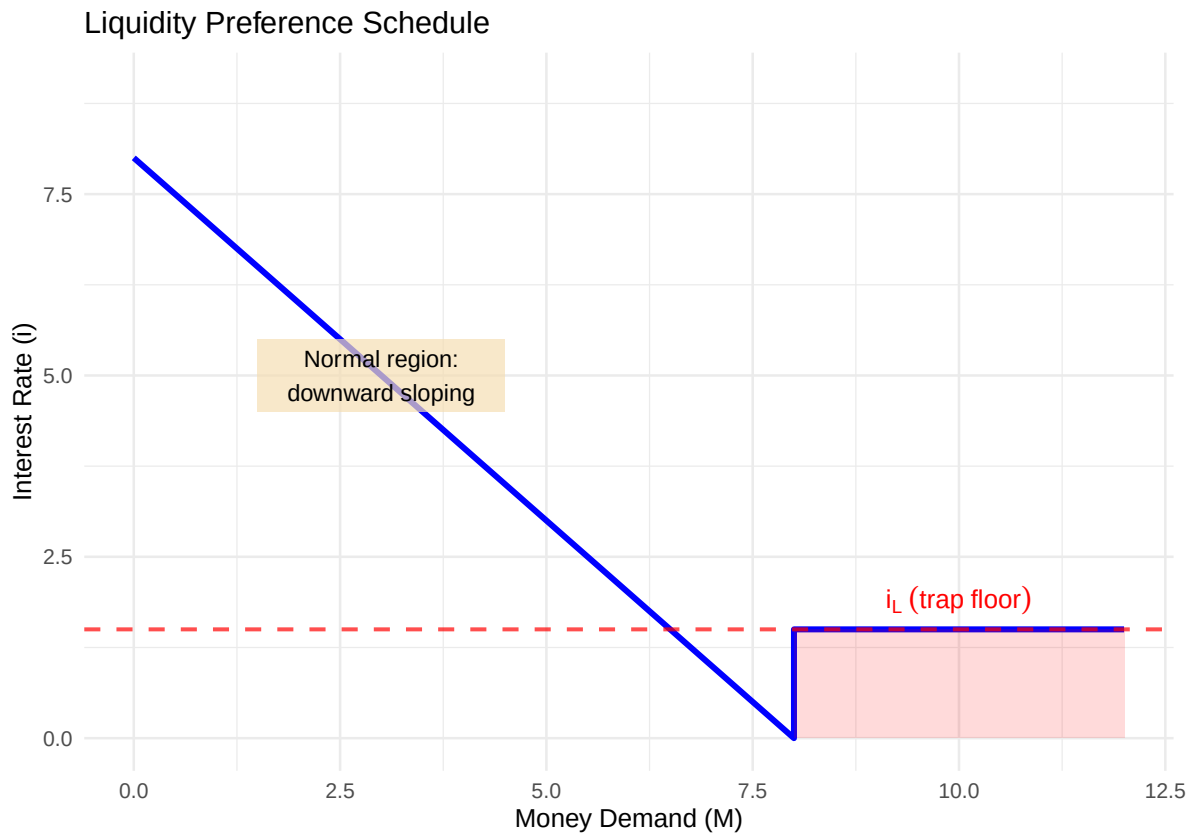


Figure 2: Liquidity preference and the liquidity trap: perfectly elastic speculative money demand at  $i$  approx  $i_L$

### 1.6.2 Implication: Monetary Expansions Lose Traction

When liquidity preference is horizontal, an increase in money supply shifts portfolios into money balances without lowering  $i$  further. This is the canonical statement that conventional monetary policy becomes ineffective in a liquidity trap. (Keynes, 1936)

### 1.6.3 Real-World Policy Responses (Historical Anchors)

The major economies have faced liquidity trap conditions: (Callen and Ostry, 2003; England, 2009; Bank, 2014)

- **Bank of England** (March 2009): Cut Bank Rate to 0.5% and launched reserve-financed asset purchases of GBP 75bn, reviewed at each meeting, to provide further monetary loosening once conventional policy was constrained. [BoE March 2009 minutes](#) and [BoE QE explainer](#). (England, 2009; Benford *et al.*, 2009)
- **Bank of Japan** (Feb 1999-Aug 2000): Adopted a Zero Interest Rate Policy (ZIRP) to stabilize activity amid deflationary pressures. [BoJ policy timeline](#). (Japan, 2023)
- **Bank of Japan** (Mar 2001-Mar 2006): Adopted Quantitative Easing Policy (QEP), expanding reserves and asset purchases after short rates had reached the floor. [BoJ policy timeline](#). (Fund, 2012; Japan, 2023)
- **European Central Bank** (June 2014-July 2022): Introduced negative deposit facility rates in June 2014 and kept them below zero until July 2022, reflecting attempts to relax constraints near the effective lower bound. [ECB June 2014 press release](#) and [ECB key interest rates table](#). (Schnabel, 2020; Bank, 2022)

These episodes are best interpreted as policy responses to Keynesian concerns about weak demand at very low rates—though modern frameworks emphasize multiple transmission channels beyond the short rate (QE, forward guidance, credit policy). (Schnabel, 2020)

COVID-19 reinforced the flight-to-liquidity logic: in March 2020 the Bank of England cut Bank Rate to 0.1% and expanded asset purchases by GBP 200bn (to GBP 645bn) to support market functioning, while the ECB announced the Pandemic Emergency Purchase Programme (PEPP) with an initial envelope of EUR 750bn to counter risks to transmission and the outlook. These actions highlight that liquidity preference can spike during stress episodes, even as policy rates approach their lower bound. [BoE March 2020 special meeting](#) and [ECB PEPP announcement](#). (England, 2020; Bank, 2020) Figure 6 shows a data-based timeline of ZLB and unconventional policy episodes across Japan, the United States, the euro area, and COVID-era interventions.

```
library(ggplot2)

timeline <- data.frame(
  policy = c("Japan ZIRP", "Japan QEP", "US ZLB", "US LSAP1", "EA negative DFR", "BoE COVID (
  start = as.Date(c("1999-02-12", "2001-03-19", "2008-12-16", "2009-03-18", "2014-06-11", "20
  end = as.Date(c("2000-08-11", "2006-03-09", "2015-12-16", "2010-03-31", "2022-07-27", "202
  y = c(7, 6, 5, 4, 3, 2, 1)
)

ggplot(timeline, aes(y = y)) +
```

```
geom_segment(aes(x = start, xend = end, yend = y), linewidth = 2, color = "steelblue") +
geom_point(aes(x = start), size = 2, color = "steelblue") +
geom_point(aes(x = end), size = 2, color = "steelblue") +
geom_text(aes(x = start, label = policy), hjust = 0, nudge_x = 30, size = 3.2) +
scale_x_date(date_breaks = "4 years", date_labels = "%Y") +
scale_y_continuous(breaks = timeline$y, labels = NULL) +
labs(x = "Year", y = NULL, title = "Timeline of ZLB and unconventional policy episodes") +
theme_minimal() +
theme(panel.grid.major.y = element_blank(),
      panel.grid.minor = element_blank())
```

Timeline of ZLB and unconventional policy episodes

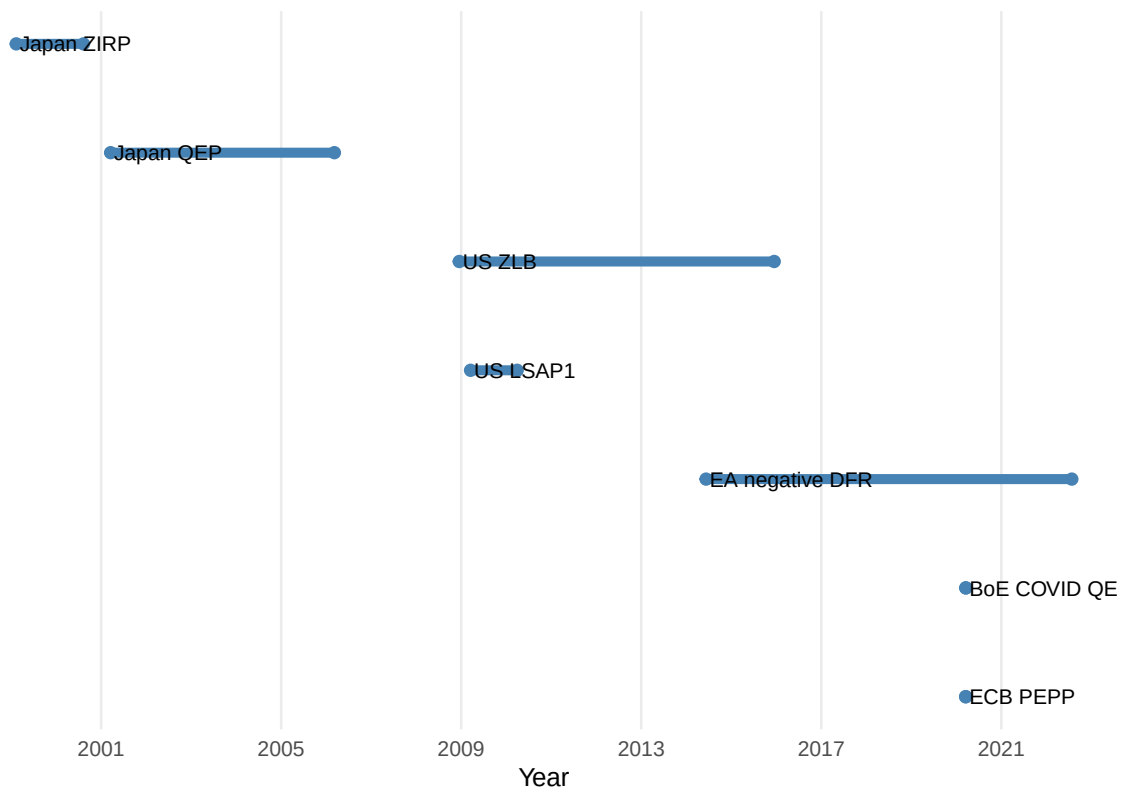


Figure 3: Timeline linking Japan (ZIRP/QE), US (ZLB + LSAP1), euro area (negative rates), and COVID-era QE/PEPP, illustrating how liquidity trap concerns reshaped central banking.

Figure 7 shows the ECB deposit facility rate path across the negative-rate period and the July 2022 exit.

```
library(ggplot2)

dfr <- data.frame(
  date = as.Date(c("2014-06-11", "2014-09-10", "2015-12-09", "2016-03-16", "2019-09-18", "2022-07-01")),
  rate = c(-0.10, -0.20, -0.30, -0.40, -0.50, 0.00)
)

ggplot(dfr, aes(x = date, y = rate)) +
  geom_step(direction = "hv", linewidth = 1.2, color = "firebrick") +
  geom_point(size = 2, color = "firebrick") +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  scale_y_continuous(breaks = c(-0.5, -0.4, -0.3, -0.2, -0.1, 0.0)) +
  labs(x = "Year", y = "Deposit facility rate (percent)", title = "ECB deposit facility rate") +
  theme_minimal()
```

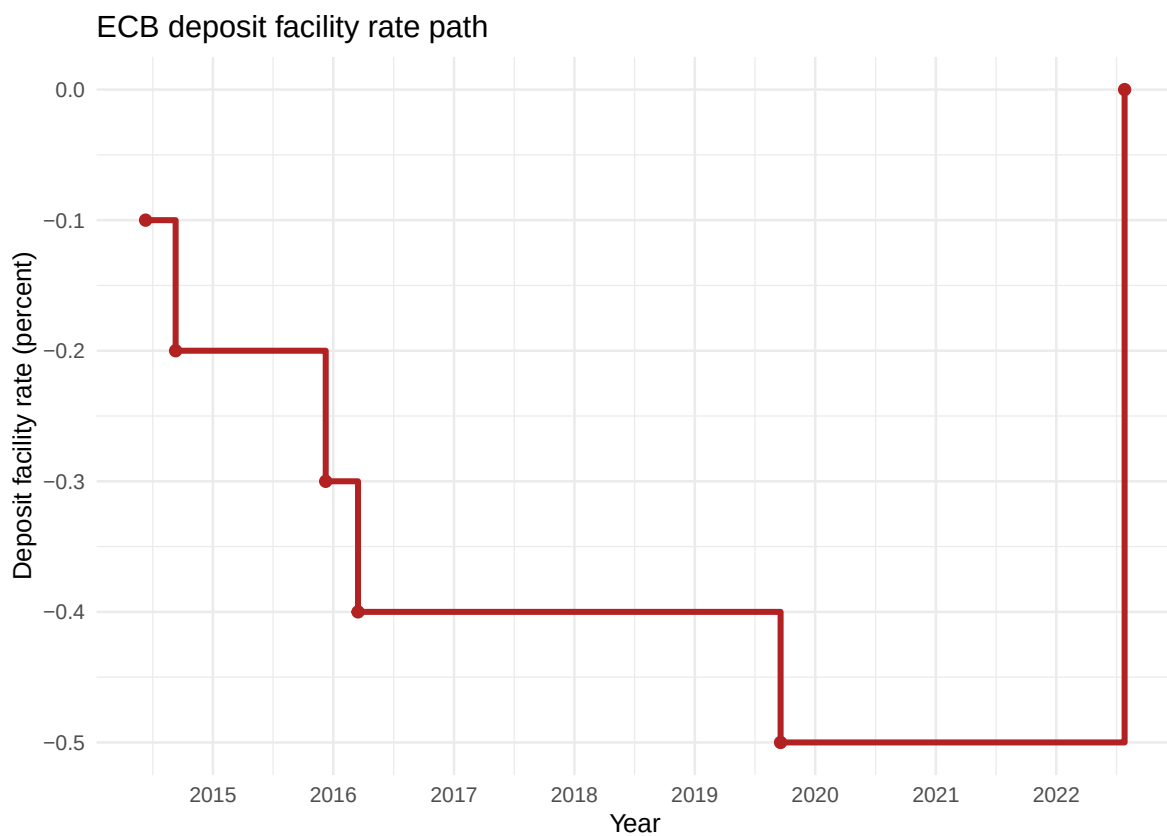


Figure 4: ECB deposit facility rate (DFR) 2014-2022, showing the move below zero and return to 0.00% in July 2022.

## 1.7 Disequilibrium Analysis Using IS-LM and AD-AS

### 1.7.1 Outside the Trap: Prices Help Restore Full Employment

In standard IS-LM, if the intersection occurs below potential output  $\bar{Y}$ , goods demand is weak, placing downward pressure on the price level  $P$ . A fall in  $P$  raises real money balances  $M/P$ , shifting LM down/right, lowering  $i$ , and increasing  $Y$ , restoring equilibrium at  $\bar{Y}$ . (Hicks, 1937; Lewis and Mizen, 2000) Figure 3 shows the deflation-driven LM shift that restores full employment outside the trap.

```
library(ggplot2)

# Output
Y <- seq(0, 10, length.out = 100)

# IS curve (downward sloping)
IS <- 8 - 0.6 * Y

# LM curves
LM1 <- 1 + 0.5 * Y # Initial LM
LM2 <- 0.3 + 0.5 * Y # LM after price fall (shifted down/right)

# Full employment output
Y_bar <- 7

# Initial equilibrium (below full employment)
Y0 <- 4.3
i0 <- 5.4

# New equilibrium (at full employment)
Y1 <- Y_bar
i1 <- 8 - 0.6 * Y1

# Plot
ggplot() +
  geom_line(aes(x = Y, y = IS), color = "blue", linewidth = 1.2) +
  geom_line(aes(x = Y, y = LM1), color = "red", linewidth = 1.2) +
  geom_line(aes(x = Y, y = LM2), color = "red", linewidth = 1.2, linetype = "dashed") +
  geom_vline(xintercept = Y_bar, linetype = "dotted", color = "gray50", linewidth = 1, alpha = 0.5) +
  annotate("point", x = Y0, y = i0, size = 4, color = "black") +
  annotate("point", x = Y1, y = i1, size = 4, color = "darkgreen") +
  annotate("segment", x = Y0, y = i0, xend = Y1, yend = i1,
```

```

        arrow = arrow(length = unit(0.3, "cm")),
        color = "darkgreen", linewidth = 1) +
annotate("text", x = Y_bar + 0.2, y = 0.5, label = "bar(Y)", parse = TRUE,
        fontface = "bold", size = 4.5) +
annotate("rect", xmin = 4.5, xmax = 6.5, ymin = 4.0, ymax = 5.0,
        fill = "lightgreen", alpha = 0.7) +
annotate("text", x = 5.5, y = 4.5,
        label = "Price falls ->\nLM shifts right",
        size = 3) +
annotate("text", x = 8.5, y = 2.5, label = "IS", color = "blue", fontface = "bold", size =
annotate("text", x = 9.5, y = 5.8, label = "LM(P[0])", parse = TRUE, color = "red", size =
annotate("text", x = 9.5, y = 5.0, label = "LM'(P' < P[0])", color = "red", size = 4) +
labs(x = "Output (Y)", y = "Interest Rate (i)",
     title = "IS-LM: Normal Times Adjustment via Deflation") +
theme_minimal() +
coord_cartesian(xlim = c(0, 10), ylim = c(0, 8))

```

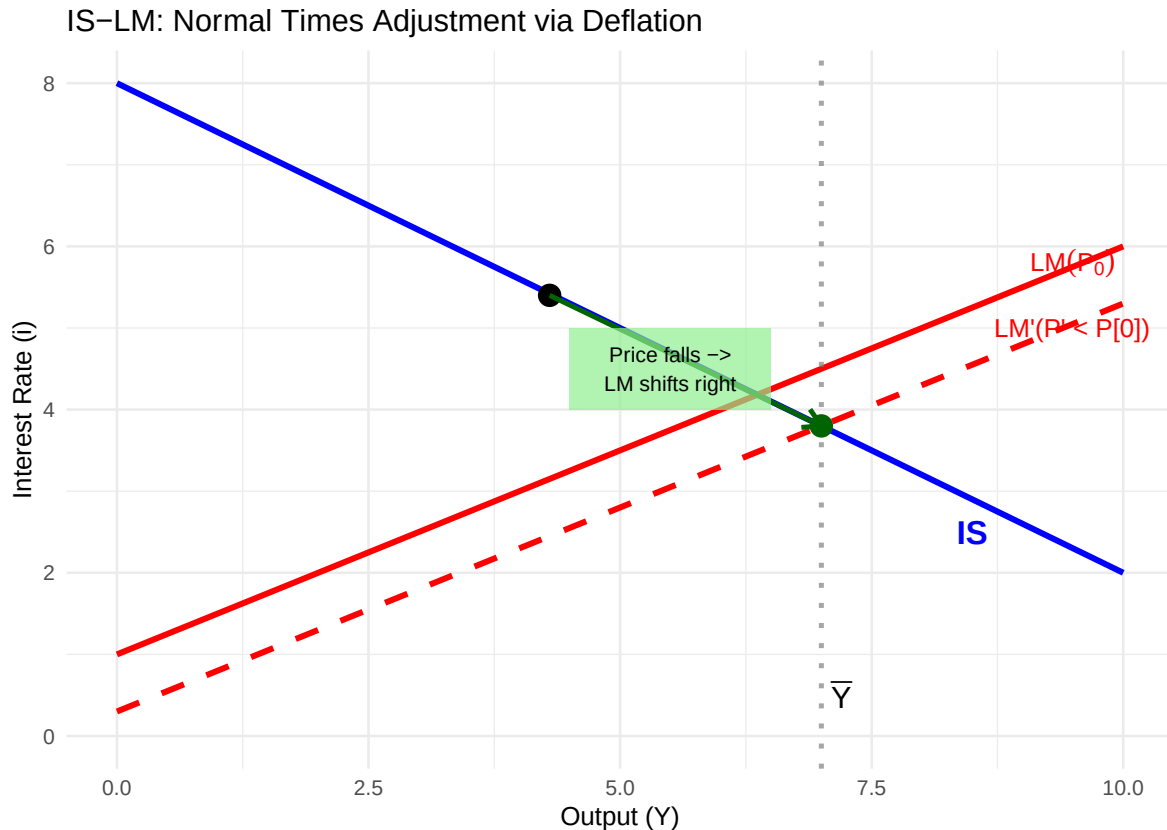


Figure 5: IS-LM adjustment in normal times: deflation shifts LM and restores full employment

**Key mechanism (Keynes effect):** Deflation  $\rightarrow$  higher real money  $\rightarrow$  lower interest  $\rightarrow$  higher investment  $\rightarrow$  higher output. ([Hicks, 1937](#))

### 1.7.2 Inside the Trap: LM Stretches Right but Does Not Shift Down

In the liquidity trap, LM is horizontal at  $i_L$ . The critical point: **falling  $P$  stretches LM rightwards but does not shift it downwards**. Therefore, the economy cannot reach  $\bar{Y}$  via the usual adjustment channel. ([Keynes, 1936](#)) Figure 4 shows the horizontal LM at the rate floor and the persistent output gap.

```
library(ggplot2)

# Output and curves
Y <- seq(0, 10, length.out = 100)
IS <- 6 - 0.5 * Y
```



```

i_L <- 2.0
Y_bar <- 7

df <- data.frame(Y = Y, IS = IS)
Y0 <- (6 - i_L) / 0.5

ggplot(df, aes(x = Y)) +
  geom_line(aes(y = IS), color = "blue", linewidth = 1.2) +
  geom_hline(yintercept = i_L, color = "red", linewidth = 1.2) +
  geom_hline(yintercept = i_L, color = "red", linewidth = 1.2, linetype = "dashed", alpha = 0.5) +
  annotate("point", x = Y0, y = i_L, color = "red", size = 4) +
  geom_vline(xintercept = Y_bar, linetype = "dotted", color = "gray50", linewidth = 1) +
  annotate("text", x = Y_bar + 0.3, y = 0.3, label = "bar(Y)", parse = TRUE, fontface = "bold", size = 12) +
  annotate("rect", xmin = min(Y0, Y_bar), xmax = max(Y0, Y_bar), ymin = 0, ymax = 5, fill = "gray50", stroke = "black", strokewidth = 1) +
  annotate("text", x = 6.5, y = 3.5, label = "Output gap: cannot close via deflation", size = 12, color = "green") +
  labs(x = "Output (Y)", y = "Interest Rate (i)", title = "IS-LM in Liquidity Trap: Interest Rate Floor") +
  theme_minimal() +
  coord_cartesian(xlim = c(0, 10), ylim = c(0, 6))

```

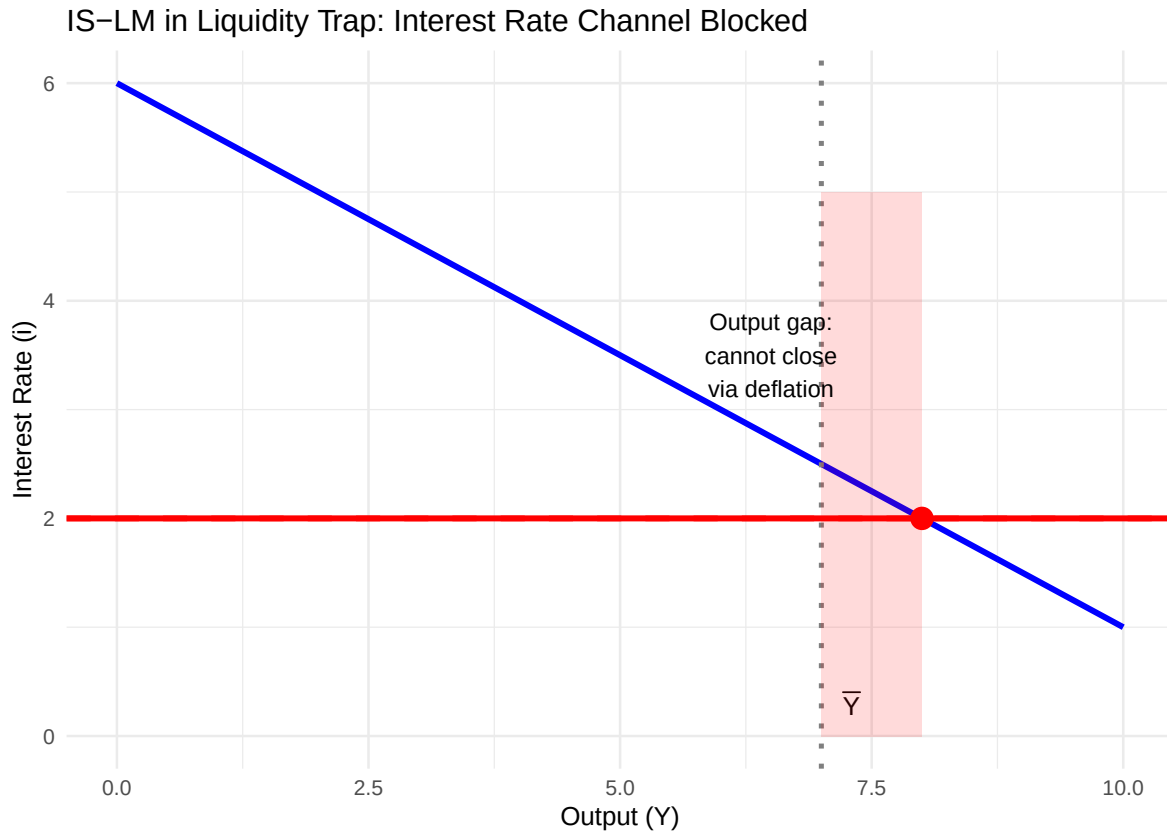


Figure 6: IS-LM in a liquidity trap: rate floor blocks adjustment

**Key insight:** Price adjustment mechanism broken; interest-rate channel blocked. ([Keynes, 1936](#))

### 1.7.3 AD Becomes (Approximately) Vertical in the Trap

Because price falls do not stimulate aggregate demand, the AD curve becomes vertical in a liquidity trap. This is the stark Keynesian disequilibrium result: even flexible prices may not restore full employment if the interest-rate mechanism is blocked. ([Keynes, 1936](#)) Figure 5 shows the vertical AD implication when the interest-rate channel is blocked.

```
library(ggplot2)

Y <- seq(0, 10, length.out = 100)
Y_bar <- 7
P_normal <- 10 - 0.8 * Y
```

```

Y_trap <- 4
P0 <- 5

df_ad <- data.frame(Y = Y, P_normal = P_normal)

ggplot(df_ad, aes(x = Y)) +
  geom_line(aes(y = P_normal), color = "blue", linewidth = 1.2, alpha = 0.5) +
  geom_vline(xintercept = Y_trap, color = "red", linewidth = 1.2) +
  geom_vline(xintercept = Y_bar, color = "green", linetype = "dashed", linewidth = 1.2) +
  annotate("point", x = Y_trap, y = P0, color = "red", size = 4) +
  annotate("text", x = Y_trap - 0.5, y = P0 + 0.5, label = "Stuck at Y", size = 3.5) +
  annotate("rect", xmin = min(Y_trap, Y_bar), xmax = max(Y_trap, Y_bar), ymin = 0, ymax = 10) +
  annotate("text", x = 5.5, y = 7, label = "Output gap", size = 4, fontface = "bold", color = "red") +
  annotate("segment", x = Y_trap, y = 8, xend = Y_trap, yend = 2,
    arrow = arrow(length = unit(0.3, "cm")), color = "red", linewidth = 1) +
  annotate("text", x = Y_trap + 0.5, y = 4.5, label = "P falls but\nY unchanged", color = "red", size = 3.5) +
  labs(x = "Output (Y)", y = "Price Level (P)", title = "AD-AS: Liquidity Trap Implies Vertical") +
  theme_minimal() +
  coord_cartesian(xlim = c(0, 10), ylim = c(0, 10))

```

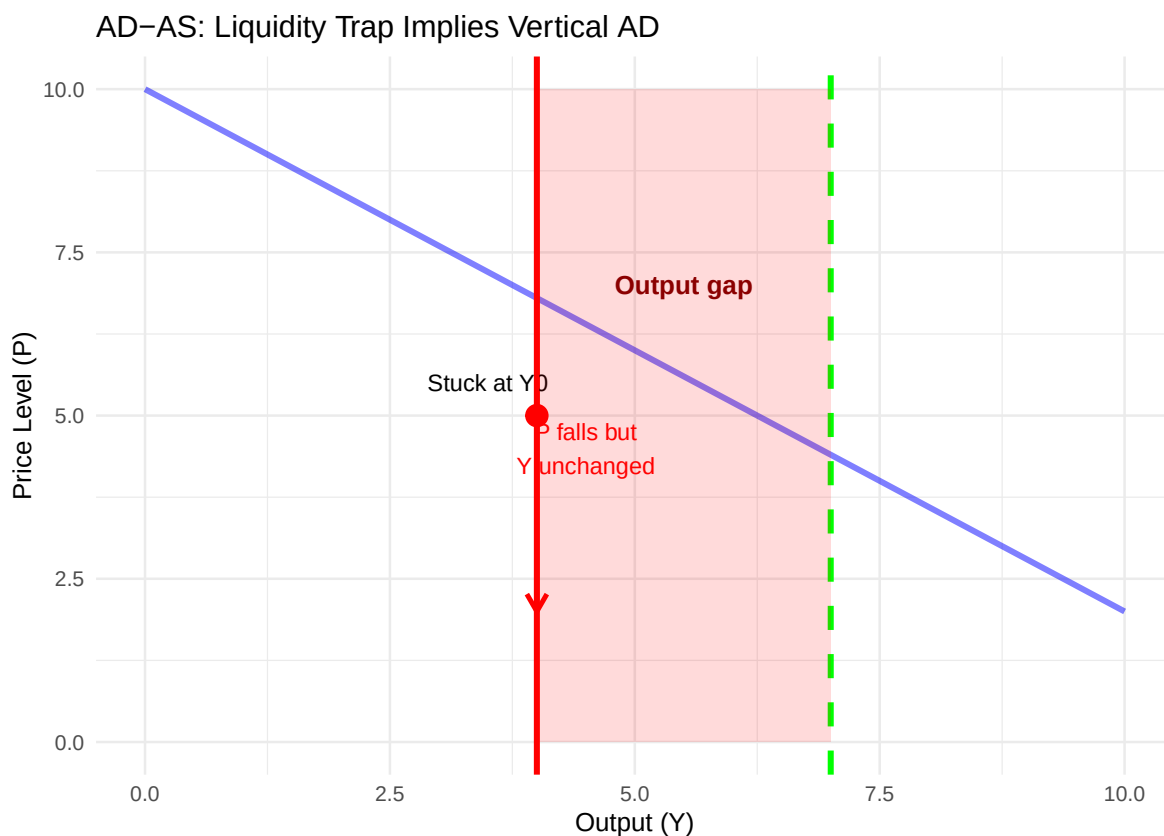


Figure 7: AD-AS in a liquidity trap: aggregate demand becomes vertical

### 1.7.4 Monetary Policy Ineffectiveness in the Trap

Any increase in money supply shifts LM rightwards but not downward, so the economy does not move to full-employment output. Hence **conventional monetary policy cannot affect aggregate demand** in a liquidity trap. This is why central banks have resorted to unconventional measures (quantitative easing, forward guidance, credit policy). ([Keynes, 1936](#); [Benford et al., 2009](#))

That is, short-rate policy is constrained, not all monetary policy: central banks can still operate through the yield curve, balance sheet, and bank lending channels. ([Schnabel, 2020](#))

The Fed's shift at the ZLB is a clean example: in December 2008 the FOMC set a 0-1/4 percent target range and explicitly pointed to balance sheet tools, followed by large-scale asset purchases in March 2009. [FOMC Dec 2008 statement](#) and [March 2009 LSAP announcement](#). ([Committee, 2008, 2009](#))

## 1.8 Neutrality and Non-Neutrality of Money

### 1.8.1 Classical View: Dichotomy & Neutrality

The **classical dichotomy** implies a separation between nominal and real variables: in equilibrium, money affects only prices, not output or employment. Money is neutral in the long run – doubling the money supply eventually doubles all prices and wages, with no impact on real GDP, employment, or real interest rates. ([Lewis and Mizen, 2000](#))

### 1.8.2 Keynes's Challenge: Uneven Adjustment & Non-Neutrality

Keynes disputed the assumption of rapid, uniform nominal price adjustment, particularly in labour markets. He observed an “unevenness” in price adjustment: some prices (like wages in long-term contracts) are quite sticky or inflexible downwards, whereas others (like commodity prices or exchange rates) might be more flexible. ([Keynes, 1936](#))

If nominal prices adjust unevenly, relative prices deviate from their market-clearing values for extended periods. Therefore: ([Keynes, 1936](#))

- **Money is not neutral** (at least over relevant horizons),
- **Dichotomy fails** because money can affect real outcomes (relative prices, unemployment),
- **Monetary policy has real effects** in the short-to-medium term due to nominal rigidities.

This line of reasoning is foundational for why modern macroeconomics treats monetary policy as potentially stabilizing in the presence of nominal rigidities—even though long-run neutrality may still hold in many models. ([Lewis and Mizen, 2000](#))

## 1.9 The Pigou Effect and Critiques of Keynes

### 1.9.1 Patinkin's Critique: Real Balances as a Quasi-Good

Don Patinkin argued that money provides “utility” because it facilitates transactions; thus money should enter utility functions as a quasi-good, particularly in real terms  $M/P$ . This insight forms the foundation of the Pigou effect critique. ([Patinkin, 1956](#)) Conceptually, Patinkin embeds real balances into a general equilibrium choice set, which is why money behaves like a quasi-good alongside other assets and goods. ([Patinkin, 1956](#))

### 1.9.2 Pigou Effect: Deflation Raises Real Balances and Shifts IS

Pigou's argument applies Patinkin's insight against Keynes's trap logic: if demand depends positively on real balances, falling prices increase  $M/P$ , potentially shifting IS right/up and closing the demand gap. (Pigou, 1943; Patinkin, 1956)

**The mechanism:** (Pigou, 1943; Patinkin, 1956) - Prices fall  $\rightarrow$  real money balances ( $M/P$ ) rise - Households and firms feel wealthier - Consumption and investment demand increase (wealth effect) - IS curve shifts rightward - Even in a liquidity trap (with  $i_L$  fixed), the rightward IS shift can eventually intersect LM at full employment  $\bar{Y}$

This implies the trap might be overcome by market forces alone. (Pigou, 1943; Patinkin, 1956)

The logical possibility is clear, but practical relevance is contested: in Japan's prolonged deflation, real-balance effects appear too weak or too slow to restore full employment quickly, especially when balance sheets are impaired and debt burdens are rising. (Callen and Ostry, 2003)

### 1.9.3 Keynesian Responses and Limitations

Keynesians offer several rebuttals: (Lewis and Mizen, 2000)

1. **Quantitatively small or very slow-acting:** In a deep recession or depression, people might be too pessimistic or liquidity-constrained to increase consumption just because prices are a bit lower. They might instead hoard any extra real wealth to rebuild balance sheets. (Keynes, 1936)
2. **Debt-deflation dynamics:** Deflation raises the real burden of debt, which can reduce debtors' spending (counteracting Pigou). This redistributes wealth from borrowers (who curtail spending) to creditors (who may not increase spending much), possibly resulting in a net drop in demand. (Lewis and Mizen, 2000)
3. **Destabilizing expectations:** Sustained deflation can worsen expectations – if people expect prices to keep falling, they might postpone purchases (the opposite of Pigou's assumption), a dynamic observed in Japan's deflationary period. (Krugman, 1998; Callen and Ostry, 2003)
4. **Institutional constraints:** Prices and wages cannot fall indefinitely due to institutional and psychological limits. Central banks actively fight deflation for these reasons—for example, the ECB's commitment to a positive inflation target (near 2%) is partly to avoid falling into a deflationary trap where any Pigou effect would be too feeble and slow. (Bank, 2014)

#### 1.9.4 Analytical Significance

All that said, the Pigou effect remains an important theoretical critique because it highlights a channel (wealth-consumption) that Keynes's original model left out. In later developments of Keynesian theory, economists like Patinkin integrated real balances into the analysis, which helped reconcile some classical and Keynesian views. They concluded that if prices are perfectly flexible and wages adjust (so no other frictions), then even a liquidity trap would eventually be resolved by sufficient deflation raising real money balances – thus restoring **long-run neutrality of money** even if short-run non-neutrality is severe. However, if there are other constraints (like the inability of nominal interest to go below zero, or institutional and psychological limits to how much and how fast prices can fall), then the economy can suffer prolonged departures from full employment – just as Keynes asserted. (Pigou, 1943; Patinkin, 1956)

#### 1.10 Summary and Key Takeaways

The following points capture the essence of Keynes's monetary economics in the context of lecture notes on Monetary Economics EC3014: (Keynes, 1936)

1. **In a monetary economy, Walras' Law does not imply Say's Law** – excess demand for money can co-exist with excess supply of goods.
2. **The money market is exceptional** – money's nominal price does not adjust, supply is exogenous, and close substitutes are limited.
3. **Usually, interest rates and/or the price level can restore equilibrium**, but sluggish adjustment can generate prolonged unemployment.
4. **In a liquidity trap, money demand is infinitely interest-elastic** – LM becomes horizontal and conventional monetary policy loses power.
5. **In that case, price changes may not raise aggregate demand** – AD becomes vertical in the trap logic.
6. **Say's Law breaks down in a monetary economy** – Keynes showed that excess saving can translate into hoarding money, destroying aggregate demand and causing unemployment.
7. **Pigou and Patinkin offer a critique** – rising real balances under deflation can potentially stimulate demand (Pigou effect), but this mechanism may be weak or slow in practice.
8. **Even outside traps, Keynes argues money is not neutral** because price adjustment is uneven; the classical dichotomy fails.

9. **Policy implications are profound** – when the interest-rate channel is blocked (liquidity trap), policymakers must rely on unconventional tools (quantitative easing, fiscal policy, forward guidance) to restore demand.

These points summarize the Keynesian framework and its policy implications. (Keynes, 1936; Lewis and Mizen, 2000)

## 1.11 Conclusion

Keynes's theory of money brought money demand and liquidity preference to the forefront of macroeconomics, fundamentally altering how we analyze interest rates and output. Unlike the classical one-dimensional view of money (for transactions only), Keynes identified multiple motives for holding money – transactions, precautionary, and speculative – which together imply that money demand depends on both income and interest rates. The interest rate, in turn, is viewed as the price that equilibrates the supply and demand for money (liquidity), rather than purely the reward for saving as in classical theory. (Keynes, 1936)

We explored the concept of the liquidity trap, Keynes's most dramatic departure from classical thinking. In a liquidity trap, extremely low interest rates lead to a horizontal money demand curve – additional money gets hoarded rather than spent. The liquidity trap highlights how an economy can get stuck with inadequate demand, zero interest rates, and even deflation, rendering conventional monetary policy ineffective. The IS-LM disequilibrium analysis showed that in such a scenario, the usual self-correcting mechanisms (interest rate adjustment and price-level adjustment) may fail to restore full employment. This underlines Keynes's broader point that money is not neutral, especially during economic downturns. (Keynes, 1936)

The experiences of Japan's lost decade and the post-2008 global slump demonstrate the importance of Keynes's insights: when money's usual role as a lubricator of the economy breaks down, proactive measures are needed to avoid long-term stagnation. At the same time, the debate between Keynesian and classical views (as epitomized by the Pigou effect discussion) highlights that the long-run interaction between money and the real economy is complex. Overall, Keynes's work ensures that economists and policymakers treat money not as a mere veil, but as a fundamental part of the economy's structure that can shape macroeconomic outcomes. (Callen and Ostry, 2003; Benford *et al.*, 2009)

## 1.12 References

Bank, E.C. (2014) "Monetary policy decisions." Press release. Available at: <https://www.ecb.europa.eu/press/pr/date/2014/html/pr140605.en.html>.

Bank, E.C. (2020) "ECB announces 750 billion euro pandemic emergency purchase programme (PEPP)." Press release. Available at: [https://www.ecb.europa.eu/press/pr/date/2020/html/ecb.pr200318\\_1~3949d6f266.en.html](https://www.ecb.europa.eu/press/pr/date/2020/html/ecb.pr200318_1~3949d6f266.en.html).



Bank, E.C. (2022) “Key ECB interest rates.” Statistical Data Warehouse table. Available at: [https://www.ecb.europa.eu/stats/policy\\_and\\_exchange\\_rates/key\\_ecb\\_interest\\_rates/html/index.en.html](https://www.ecb.europa.eu/stats/policy_and_exchange_rates/key_ecb_interest_rates/html/index.en.html).

Benford, J. *et al.* (2009) “Quantitative easing,” *Bank of England Quarterly Bulletin* [Preprint]. Available at: <https://www.bankofengland.co.uk/quarterly-bulletin/2009/q2/quantitative-easing>.

Callen, T. and Ostry, J.D. (2003) *Japan’s lost decade: Policies for economic revival*. Washington, DC: International Monetary Fund. Available at: <https://www.elibrary.imf.org/view/book/9781589062330/9781589062330.xml>.

Committee, F.O.M. (2008) “Federal reserve issues FOMC statement.” Press release. Available at: <https://www.federalreserve.gov/newsevents/pressreleases/monetary20081216b.htm>.

Committee, F.O.M. (2009) “Federal reserve announces expanded asset purchases.” Press release. Available at: <https://www.federalreserve.gov/newsevents/pressreleases/monetary20090318a.htm>.

England, B. of (2009) “MPC announcement: Bank rate reduced to 0.5.” Press release. Available at: <https://www.bankofengland.co.uk/news/2009/march/monetary-policy-committee-reduces-bank-rate-to-05-and-initiates-asset-purchases>.

England, B. of (2020) “Monetary policy committee announces additional measures to support the economy.” Press release. Available at: <https://www.bankofengland.co.uk/news/2020/march/monetary-policy-committee-announces-additional-measures-to-support-the-economy>.

Fund, I.M. (2012) *Japan: 2012 article IV consultation - staff report; public information notice on the executive board discussion; and statement by the executive director for japan*. IMF Country Report 12/208. International Monetary Fund. Available at: <https://www.imf.org/en/Publications/CR/Issues/2016/12/31/Japan-2012-Article-IV-Consultation-Staff-Report-Public-Information-Notice-on-32413>.

Hicks, J.R. (1937) “Mr. Keynes and the ”classics”: A suggested interpretation,” *Econometrica*, 5(2), pp. 147–159. Available at: <https://www.jstor.org/stable/1907242>.

Japan, B. of (2023) “Monetary policy measures in japan: History and chronology.” Timeline. Available at: <https://www.boj.or.jp/en/mopo/outline/chrono/index.htm/>.

Keynes, J.M. (1936) *The general theory of employment, interest and money*. London: Macmillan.

Krugman, P. (1998) “It’s baaack: Japan’s slump and the return of the liquidity trap,” *Brook-*

*ings Papers on Economic Activity*, (2), pp. 137–205. Available at: <https://www.brookings.edu/bpea-articles/its-baaack-japans-slump-and-the-return-of-the-liquidity-trap/>.

Lewis, M.K. and Mizen, P.D. (2000) *Monetary economics*. Oxford: Oxford University Press.

Patinkin, D. (1956) *Money, interest, and prices*. Evanston, IL: Row, Peterson; Company.

Pigou, A.C. (1943) “The classical stationary state,” *The Economic Journal*, 53(212), pp. 343–351. Available at: <https://www.jstor.org/stable/2226322>.

Schnabel, I. (2020) “Going negative: The ECB’s experience.” European Central Bank; Speech, Jackson Hole Economic Symposium. Available at: <https://www.ecb.europa.eu/press/key/date/2020/html/ecb.sp200828~3e197db7e1.en.html>.

Walras, L. (1874) *Éléments d’économie politique pure (elements of pure economics)*. Lausanne.